Def. Let $p: E \to B$ be a continuous onto map.
An open set $U \subset B$ is said to be evenly covered by p if:

$$p^{-1}[U] = \bigcup_{\alpha \in I} V_\alpha \quad \text{where } V_\alpha \text{ is dissjoint open sets such that}$$

$$\text{for all } \alpha \in I,\ U \overset{p}{\cong} V_\alpha$$

The map $p: E \to B$ is called a Covering map if:

for each $b \in B$, there exists an evenly covered neibor U containing b.

If such a map p exists, then E is called the covering space.

Prop. Every covering map is open map.
Proof. Let $p: E \to B$ be a covering map, and $W \subset E$ be an open set.
To show $p[W]$ is an open set, let $b \in p[W]$ be given.
Since the p is covering map, there exists an evenly covered open set U containing b. By definition, for some dissjoint open sets V_α ($\alpha \in I$),

$$p^{-1}[U] = \bigcup_{\alpha \in I} V_\alpha \qquad b \in U$$

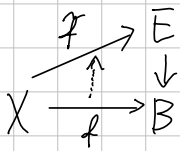
and, $b \in p[W]$, so $b = p(y)$ for some $y \in W$. Since $b \in U$, so $p^{-1}[p[\{y\}]] \in p^{-1}[U]$,
therefore $y \in p^{-1}[U] = \bigcup_{\alpha \in I} V_\alpha$, thus $y \in V_{\alpha_0}$ for some $\alpha_0 \in I$.
Now, $V_{\alpha_0} \cap W$ is an open set of E containing y, so that
$V_{\alpha_0} \cap W$ is an open set of V_{α_0}.
Since the restriction

$$p|_{V_{\alpha_0}}: V_{\alpha_0} \to U$$

is Homeomorphism, so $p[V_{\alpha_0} \cap W] \subset U$ is an open set of U,
and since U is open set of B, thus $p[V_{\alpha_0} \cap W]$ is open set of B.
Finally, $b = p(y) \in p[V_{\alpha_0} \cap W] \subset p[W]$, so $p[W]$ is an open set of B. ∎

Lift.

Def. Let $p: E \to B$ be a continuous map.
For a continuous map $f: X \to B$, a lifting of f is;
a continuous map $\tilde{f}: X \to E$ such that $p \circ \tilde{f} = f$.



Lemma. Let $p: E \to B$ be a covering map and $f: X \to B$ be continuous maps.
If there exist two lifts $\tilde{f}_1, \tilde{f}_2: X \to E$ of f with $\tilde{f}_1(x_0) = \tilde{f}_2(x_0)$ for some $x_0 \in X$,
and X is connected, then $\tilde{f}_1 = \tilde{f}_2$

Thm. Let $p: (E, e_0) \to (B, b_0)$ be a covering map.
For any path $\alpha: I \to B$ such that $\alpha(0) = b_0$, there exists a unique path
$\tilde{\alpha}: I \to E$ such that $p \circ \tilde{\alpha} = \alpha$ and $\tilde{\alpha}(0) = e_0$.

Diagrammically,

